A.I Fundamentals

Module 5 Study Guide

Automated Trading & Risk Analysis

This guide covers every lesson of the module, using only words and figures from the lessons themselves. Work through it after watching the lessons, then use the checklist at the end to test yourself.

Lesson 1: Algorithmic and Automated Trading Explained

In this lesson: Defining Trading Bots; If-Then Rules Explained; Human Discretion vs Automation

Key points

- Algorithmic trading executes predefined rules automatically without emotional intervention
- Stop-loss rules manage downside risk like in our TT$1,000 example
- Automation provides speed and discipline rather than profit guarantees

Examples from the lesson

- We are stepping into Module 5, titled Automated Trading and Risk Analysis, and we are kicking things off with Lesson 1: Algorithmic and Automated Trading Explained.
- You tell yourself, if sweet peppers are less than TT$10 a pound, buy five pounds, but if they cost more, buy nothing and walk away.
- Suppose you set up a simple trading strategy with a starting capital of TT$1,000.
- You configure your software with a strict rule: purchase a specific asset if its unit price drops to TT$200, and immediately set a protective stop-loss rule to sell if the price drops further to TT$180 to limit potential loss.
- When the market price hits TT$200, the system automatically triggers an order buying 5 units, spending your entire TT$1,000 allocation.
- Now, if market conditions shift unfavorably and the asset price falls to your pre-set limit of TT$180, the bot executes the stop-loss order without delay.
- Selling those 5 units at TT$180 brings back TT$900 in total cash.
- When you evaluate this completed transaction, your gross return from the sale is TT$900.
- Comparing that back to your initial capital of TT$1,000, your final account balance reflects a net loss of TT$100.
- The bot did not save you from a small loss, but by executing the rule disciplinedly, it prevented a much larger drop from wiping out your entire TT$1,000 capital.
- A trading bot will follow a bad strategy just as faithfully as a good one.
- We saw through our TT$1,000 example how automated stop-loss rules limit downside risk, and we emphasized that automation is a tool for rule enforcement rather than a promise of profit.

Lesson 2: AI for Risk Analysis

In this lesson: Define AI-supported risk analysis workflow; Gather and prepare portfolio data; F

Key points

- Gather data & run AI models
- Calculate baseline VaR ≈ TT$2,244
- Stress-test to VaR ≈ TT$4,800 & uncover bank concentration

Examples from the lesson

- This is Module 5, Automated Trading & Risk Analysis, Lesson 2: AI for Risk Analysis.
- Monte-Carlo simulation then generates thousands of possible future price paths, allowing us to stress-test the portfolio under different scenarios. Step one is to gather your data. Suppose you have a TT$100,000 portfolio made up of TT$40,000 in ANSA Bank shares, TT$30,000 in Republic Bank shares, TT$20,000 in a fixed-deposit at your local credit union, and TT$10,000 in a small company that provides decorations and sound systems for Carnival fetes.
- Step two is to calculate baseline risk metrics with AI.
- The model estimates the daily volatility of ANSA Bank at 1.2%, Republic Bank at 1.0%, the credit union deposit at 0.2% (since it's a fixed return), and the Carnival supplier at 2.5%.
- Using historical correlations—ANS A and Republic share a 0.6 correlation, both have a 0.3 correlation with the credit union, and the supplier shows only a 0.1 correlation with the banks—the AI computes the portfolio's overall variance.
- Now for the worked example: the portfolio variance works out to 0.000184, giving a portfolio standard deviation of about 1.36% per day.
- At a 95% confidence level, the one-day Value-at-Risk (VaR) equals roughly 1.65 times the standard deviation times the portfolio value, which is TT$100,000 × 0.0136 × 1.65 ≈ TT$2,244.
- This means there is a 5% chance the portfolio could lose more than TT$2,244 in a single day under normal market conditions. Step three is scenario testing. The AI runs a Monte-Carlo simulation where oil prices drop 20%, a major fete is cancelled due to rain, and the credit union faces a temporary liquidity squeeze.
- In these stressed paths, the portfolio's VaR jumps to about TT$4,800, showing how concentrated risks can amplify losses when multiple stressors hit together.
- Although the portfolio looks diversified across four assets, the model flags that the two bank holdings together contribute over 70% of the total risk because of their high correlation and similar exposure to energy-linked earnings.
- So let's recap: we gathered portfolio data, used AI to estimate volatilities and correlations, calculated a baseline VaR of about TT$2,244, stress-tested the portfolio to see VaR rise to roughly TT$4,800 under adverse scenarios, and uncovered that the bank stocks drive most of the risk despite apparent diversification.

Lesson 3: Limits, Ethics, and Human Oversight

In this lesson: Understand flash-crash risks; Recognise model failure; Set trading limits and ch

Key points

- Flash crashes can amplify sudden market movements
- AI models can fail when conditions change
- Risk limits, ethical checks, and human checkpoints preserve accountability

Examples from the lesson

- This is A.I Fundamentals, Module 5, Automated Trading and Risk Analysis, Lesson 3: Limits, Ethics, and Human Oversight.
- Suppose you have TT$20,000 allocated to a trading account, and your personal trading plan limits the amount exposed to one trade to two percent of that account. Two percent of TT$20,000 is TT$400. That TT$400 is your planned maximum loss for the trade, not a target and certainly not a guaranteed outcome.
- However, during a sudden price drop, the market may move past your stop price before an order can execute. You could lose more than TT$400. This is why a risk limit is a control, not a promise that losses cannot exceed it.

Recap checklist — can you say each one?

Lesson 1: Algorithmic and Automated Trading Explained

☐ Algorithmic trading executes predefined rules automatically without emotional intervention

☐ Stop-loss rules manage downside risk like in our TT$1,000 example

☐ Automation provides speed and discipline rather than profit guarantees

Lesson 2: AI for Risk Analysis

☐ Gather data & run AI models

☐ Calculate baseline VaR ≈ TT$2,244

☐ Stress-test to VaR ≈ TT$4,800 & uncover bank concentration

Lesson 3: Limits, Ethics, and Human Oversight

☐ Flash crashes can amplify sudden market movements

☐ AI models can fail when conditions change

☐ Risk limits, ethical checks, and human checkpoints preserve accountability